Introduction to Database

INT191 Relational Database Concepts and Design

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What we will learn

* SQL Review
* Database Concept
* Components in Database System
* Database Management System (DBMS)
* Actors in the Database Environment

SQL - Database Language

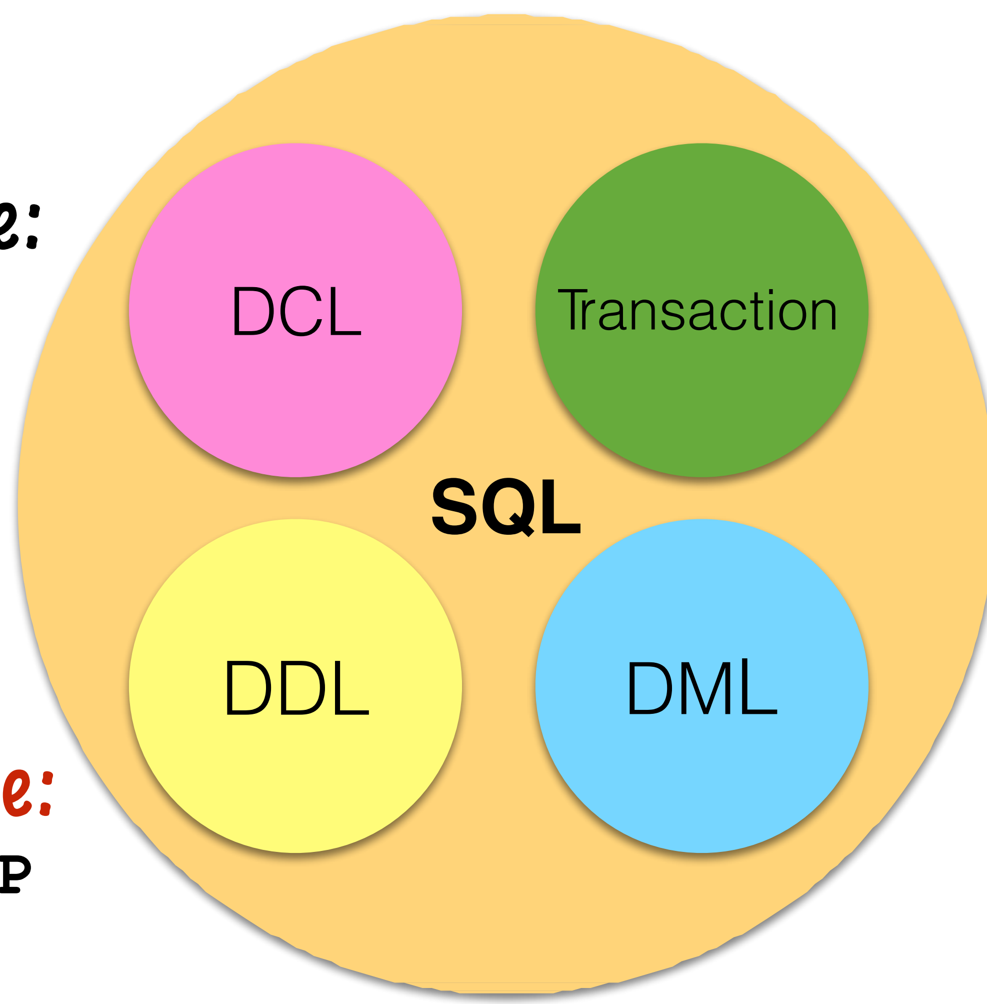
* Originally: Structured English Query Language (SEQUEL)
* Officially: Structured Query Language (**SQL**)=> Query Language

Data Control Language:

- GRANT; REVOKE

Transaction Control:

- COMMIT; ROLLBACK;
- SAVEPOINT;
- SET ISOLATION



Data Definition Language:

- CREATE; ALTER; DROP

Data Manipulation Language:

- SELECT; INSERT;
- UPDATE; DELETE;

DML Statement

```
SELECT [DISTINCT] { * | column | expression [[AS] alias][,...]}
FROM  table1 [,table2, table3,...]
 [WHERE condition(s)]
 [GROUP BY  columnList]
 [HAVING aggregate_condition]
 [ORDER BY {column|expr|alias} [ASC|DESC][,...]] ;
```

- Aggregate functions are **COUNT, COUNT DISTINCT, MIN, MAX, SUM, AVG, …**
- SET operations: **UNION, UNION ALL, EXCEPT/MINUS, INTERSECT**
- JOIN: **INNER JOIN, SELF JOIN, OUTER JOIN**
- JOIN Syntax: **JOIN..ON, JOIN..USING**

DML Statement

```
INSERT INTO  table_name (column1, column2,…, columnN)
VALUES  ( value1, value2,…, valueN);

UPDATE table_name
SET  column1 = value1 [,column2 = value2,…]
[WHERE condition(s)];

DELETE [FROM] table_name
[WHERE condition(s)];

COMMIT; / ROLLBACK;
```

Models of Data

Structured data	Semi-structured data	Unstructured data
Databases	XML / JSON data	Audio
	Email	Video
	Web pages	Image data
		Natural language
		Documents

Jones, M. Tim. 2018. "Data, structure, and the data science pipeline." IBM Developer, February 01. Accessed 2019-10-20.

RDBMS

NoSQL Database

https://www.pexels.com/photo/items-organized-on-shelves-3687999/

Purchases from the supermarket

When you **purchase goods** from your local supermarket, it is likely that a **database is accessed**. The checkout assistant uses a ***bar code reader to scan*** each of your purchases. This reader is linked to a database application that uses the bar code to ***find out the price*** of the item from a product database. The application then reduces the number of such items in stock and displays the price on the cash register.

If the reorder level falls below a specified threshold, the database system may ***automatically place an order*** to obtain more of that item. If a customer telephones ***the supermarket, an assistant can check whether an item is in stock*** *by running an application program that determines availability from the database.*

A Database

* What is a database?
* Why do we need the database?
* How does the database system work?

Database Definition

"A **shared** collection of logically **related data** (and a description of this data), designed to meet the information needs of an organization."

"A collection of information that is organized so that it can **easily be accessed, managed and updated**."

"A collection of **persistent** data that can be **shared and interrelated**."

Why do we need the database?

Goal of Database System

* To provide a way to store and retrieve database information conveniently and efficiently.
* DBMSs must meet the following requirements:
    * **Data Persistency** : the data must outlast their creators
    * **System Reliability** : recover correctly and promptly, if crash
    * **Scalability** : handle large numbers of data and lots of concurrent clients/users

Common Characteristics

- **Common Characteristics** in Database Systems
  - **Self-Describing Nature**
    - A DBMS catalog stores the description of a particular database. (e.g. data structures, types, and constraints)
    - The description is called **meta-data***.
  - **Insulation between programs and data**
    - Called program-data independence.
    - Allows changing data structures and storage organization without having to change the DBMS access programs.

Common Characteristics

- **Common Characteristics** in Database Systems
  - **Data Abstraction**
    - A **data model** is used to hide storage details and present the users with a conceptual view of the database.
    - Programs refer to the data model constructs rather than data storage details.
  - **Data Integrity**
    - Constraint: Primary Key, Unique Key, Foreign Key, etc.
  - **Data Security**
    - Control Access
    - Authorization

Common Characteristics

- **Common Characteristics** in Database Systems
  - **Multiple Views**
    - Each user may see a different view of the database, which describes only the data of interest to that user.
  - **Sharing of data and multi-user transaction processing**
    - Allowing a set of concurrent users to retrieve from and to update the database.
    - **Concurrency control** within the DBMS guarantees that each transaction is correctly executed or aborted
    - **Recovery subsystem** ensures each completed transaction has its effect permanently recorded in the database

Database Concepts

* A Database System (DBS) consists of
  - A Database (DB) and
  - A Database Management System (DBMS)
* A Database is
  - a collection of well-organized and interrelated data
* A Database Management System is
  - a set of programs to manipulate those data

Simplified Database System Environment

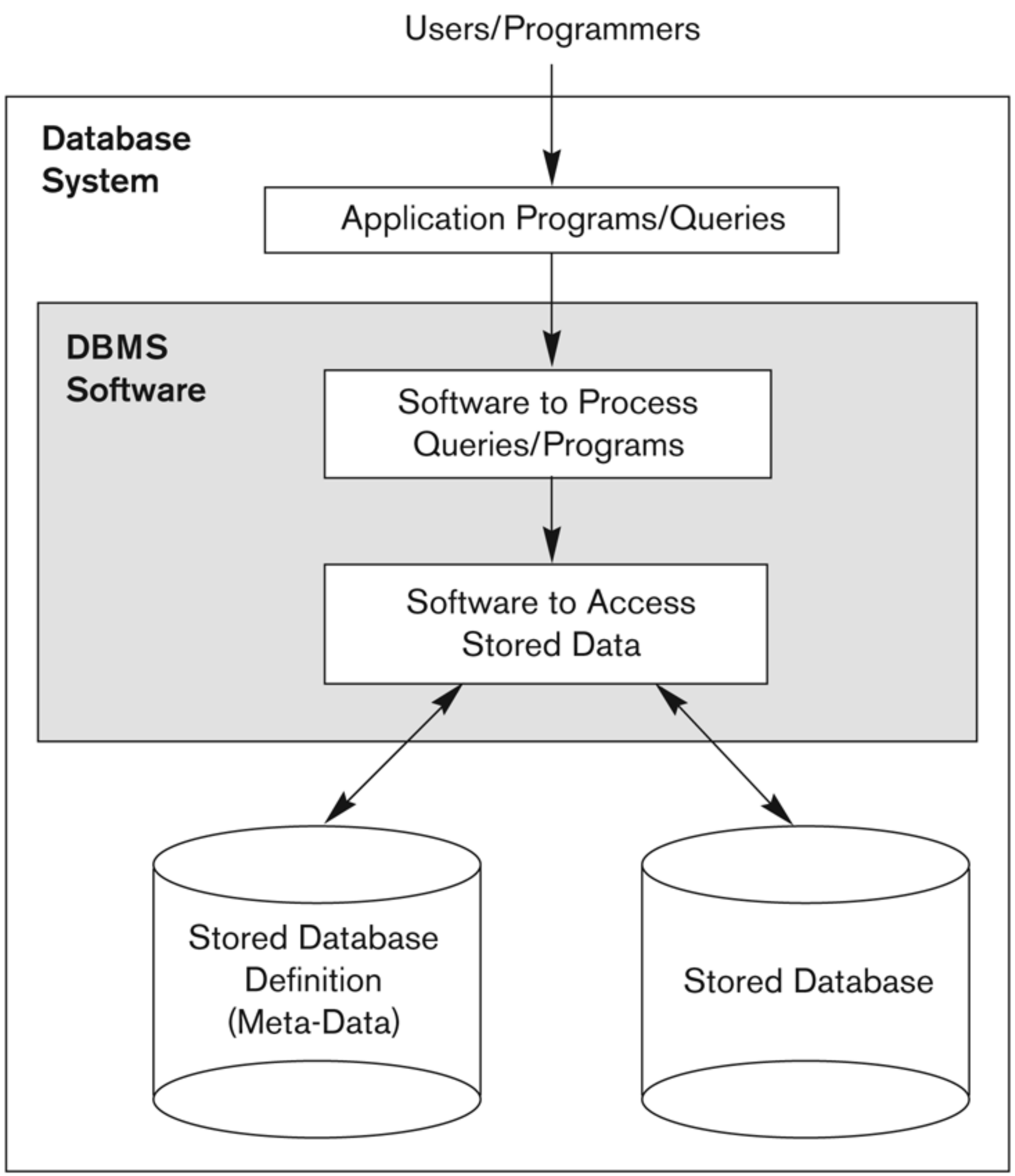
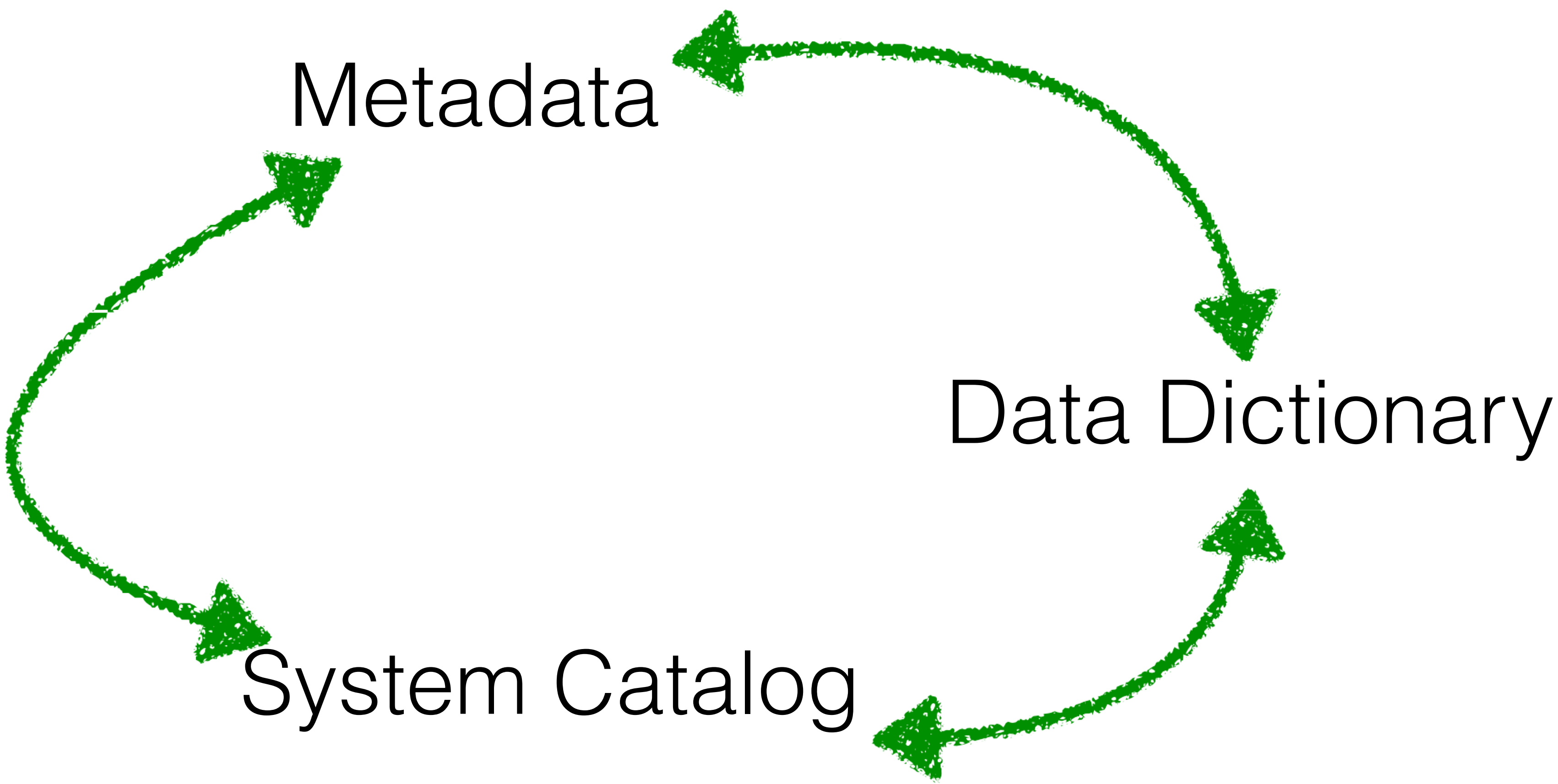


Figure 1.1
A simplified database system environment.

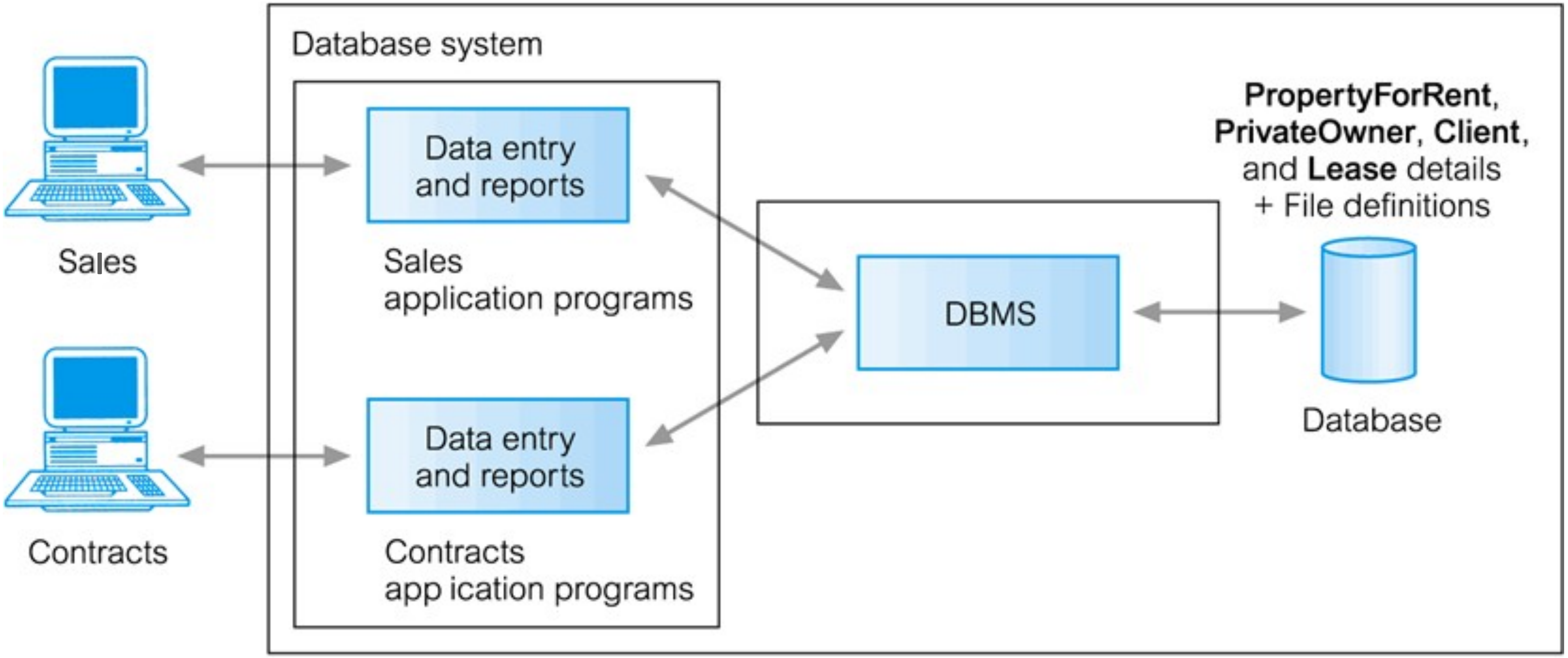
Meta-Data

“The description of the data”



https://dataedo.com/blog/what-is-metadata-examples

Database System



PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentF nish)

Advantages of Using Database System

- Data usage
  - Sharing of data
  - More information from the same amount of data
- Data management
  - Control of data redundancy
  - Data consistency
  - Improved data integrity
  - Increased concurrency
- Business operation
  - Economy of scale
  - Data independence
  - Improved security
  - Improved backup and recovery services

Relational Databases

* **(Single) DBMS - Database Server**
  * A DBMS consist of multiple databases
  * Each Database consists of
    * a set of **metadata** (information about data structures in the database)
    * A set of **tables**, views, users, triggers, indices, etc.
    * Each table consists of multiple **rows**
    * Each row consists of multiple **columns**
    * Usually, one data item is equivalent to one row of data in a table

Tables

"A named, two dimensional arrangement of data that consists of a heading part and a body part"

Columns/Attributes/Fields

PropertyForRent

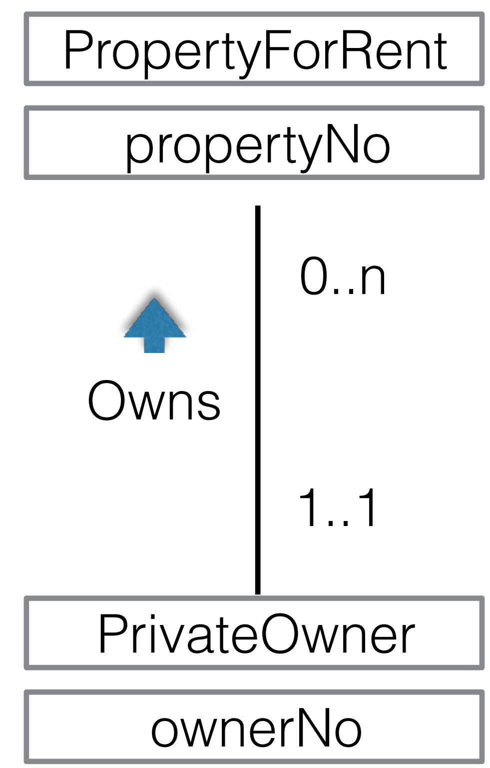
propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93

Rows/Records

Relationship

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025



Database Management System (DBMS)

"A software system that enables users to define, create, maintain, and control access to the database"

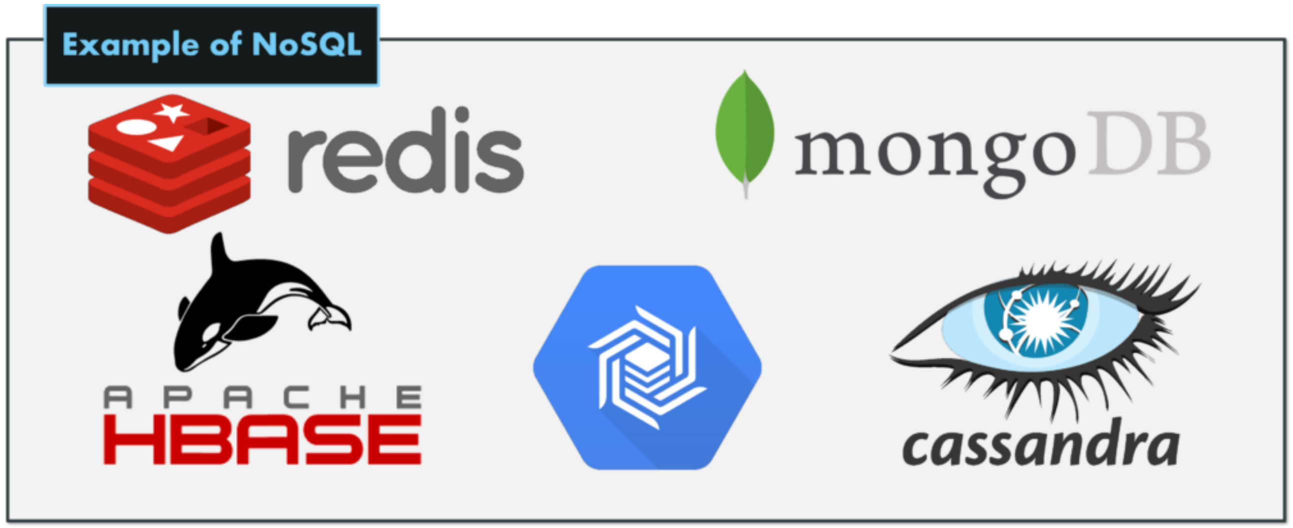
SQL

Application programs ⟷ DBMS ⟷ Database

DBMS

Relational DBMS

NoSQL Database

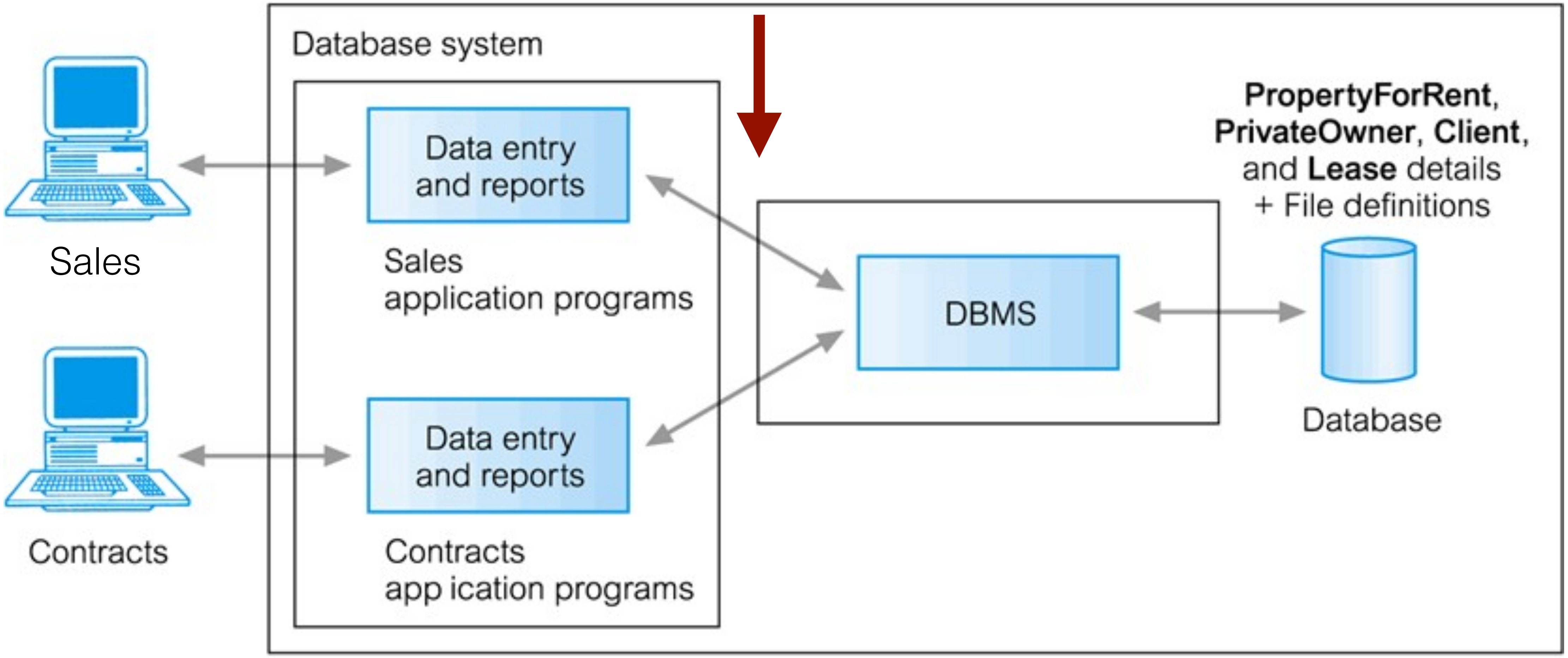


edureka!

https://www.edureka.co/blog/sql-vs-nosql-db/

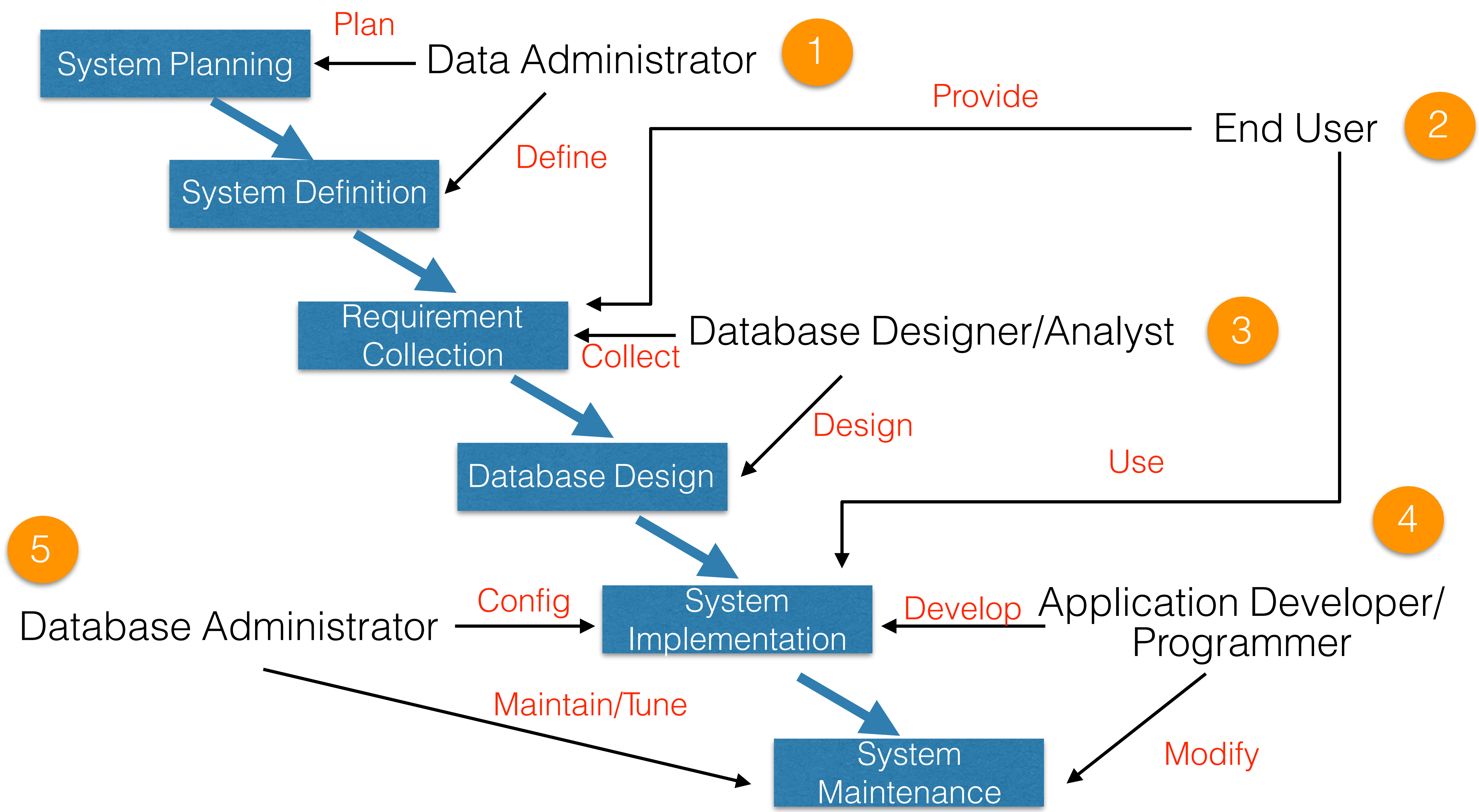
Sample: Database System

```
SELECT propertyno, type, rooms, rent
FROM   propertyforrent
WHERE  propertyno = 'PA14'
```

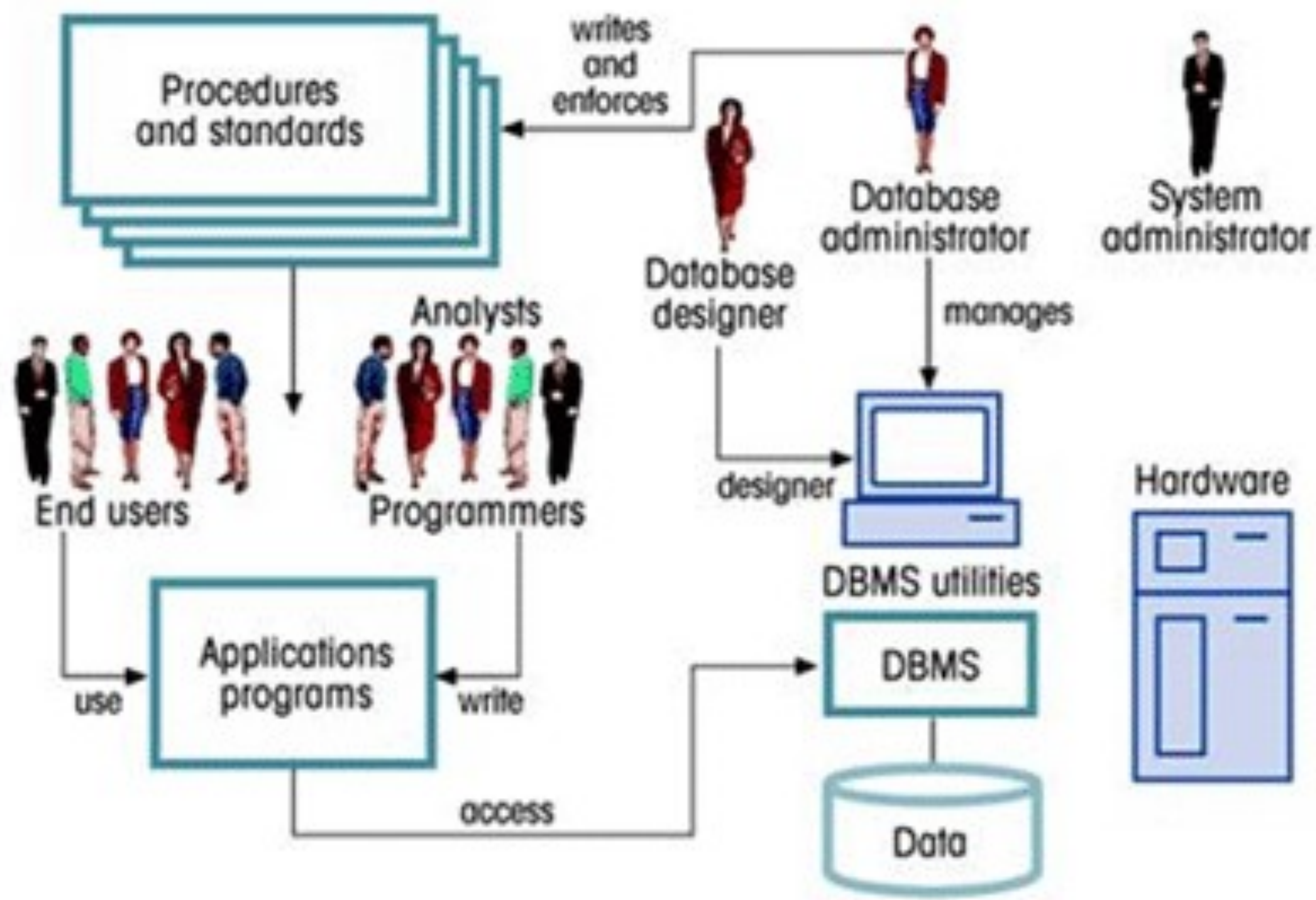


PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)
PrivateOwner (ownerNo, fName, lName, address, telNo)
Client (clientNo, fName, lName, address, telNo, prefType, maxRent)
Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentF nish)

Actors in the Database Environment



Database System Environment



THE DATABASE SYSTEM ENVIRONMENT

Source: http://dbms-ii.blogspot.com/p/dbms-architecture.html

References

* Database Design, Application Development, and Administration, Michael V Mannio (Third edition), McGrall Hill.

* Database Systems A Practical Approach to Design, Implementation and Management, Thomas Connolly, Carrolyn Begg, (Fifth edition), Addison Wesley.

* Fundamentals of Database Systems, Elmasri, Navaho (Seventh edition), Pearson.